Title

Machine learning based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

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Date: March 2026

2.Abstract

This project utilizes Machine Learning to enhance real estate market intelligence through automated buyer segmentation and investment profiling. By applying the K-Means Clustering algorithm to global property data, buyers are categorized into Premium, Budget, and Investor segments. The study features an interactive Streamlit dashboard that performs Exploratory Data Analysis (EDA) to visualize key correlations between geographic distribution, financial loan status, and acquisition intent. The final model transforms complex datasets into actionable strategic insights, providing a data-driven framework for identifying high-value market trends and consumer behaviors.

3.Keywords

Real Estate Analytics, Buyer Segmentation, Machine Learning, K-Means Clustering, Streamlit Dashboard, Data Analysis, Investment Profiling.

4.Introduction

The real estate market is increasingly driven by data-centric strategies. This study, "Real Estate AI Insights," implements Machine Learning to decode buyer behavior and global investment trends. By leveraging the K-Means Clustering algorithm, the project segments market participants into distinct groups based on their financial profiles and buying intent.

The research features a Streamlit-based dashboard (deployed via GitHub) for real-time Exploratory Data Analysis (EDA). It visualizes the correlation between geographic location, loan dependency, and acquisition purpose. This tool transforms raw datasets into actionable intelligence, providing a strategic framework for optimized marketing and investment decision-making.

5.Problem Statement

The real estate industry struggles to accurately segment buyers due to the high volume and complexity of transactional data. Traditional analysis methods are often static and fail to capture the dynamic relationship between a buyer's geographic location, financial profile, and investment intent.

Without automated insights, it is difficult to differentiate between Premium, Budget-Conscious, and Investor personas.

6.Methodology

The research follows a structured data science pipeline to transform raw real estate data into actionable market intelligence. The methodology is divided into the following key phases:

- Data Collection & Preprocessing: The process begins with sourcing a global real estate transaction dataset. Data cleaning is performed using Python (Pandas) to handle missing values, normalize financial figures, and encode categorical variables such as "Loan Status" and "Acquisition Purpose."
- Exploratory Data Analysis (EDA): A comprehensive EDA is conducted to identify underlying patterns. This involves univariate and bivariate analysis to understand the distribution of buyer ages, geographic concentrations, and the relationship between client categories and their strategic intent.
- Machine Learning Clustering: To achieve automated buyer segmentation, the K-Means Clustering algorithm is implemented. Features such as investment amount, loan dependency, and geographic data are used to group participants into three distinct personas: Premium, Budget-Conscious, and Investor.
- Dashboard Development & Deployment: An interactive web interface is built using the Streamlit framework. High-fidelity visualizations, including 3D Orthographic Globes and Sunburst Charts, are integrated using Plotly Express. The final application is version-controlled via GitHub and deployed on Streamlit Cloud for real-time accessibility.

7.System Architecture

The architecture of the Real Estate AI Insights system is designed as a modular, cloud-deployed analytical pipeline. It follows a four-tier structure to ensure seamless data flow and real-time visualization:

- Data Source Layer: The foundation consists of raw real estate transaction data stored in CSV format (Cleaned_Real_Estate_Market_Intelligence.csv). This layer

provides the raw features such as buyer age, loan status, and geographic coordinates.

- Processing & Analytics Layer (Python): This is the core engine where Pandas and NumPy handle data cleaning and transformation. The K-Means Clustering algorithm is implemented here to process multi-dimensional features and generate buyer segments (Premium, Budget, Investor).
- Visualization & Interface Layer (Streamlit & Plotly): The frontend is built using the Streamlit framework. It integrates Plotly Express to render interactive 3D orthographic globes, sunburst strategic intent charts, and segmented histograms.
- Deployment & Cloud Layer (GitHub & Streamlit Cloud): The entire codebase is version-controlled via GitHub. Continuous Deployment (CD) is achieved through Streamlit Cloud, which pulls the app.py and requirements.txt files to host the application live on a public URL.

Tip for your Document:

CSV Data ➤ Python (Clustering) ➤ Streamlit (UI) ➤ GitHub/Cloud (Deployment).

8.Results & Key Insights

The K-Means Clustering algorithm successfully segmented the real estate market into three distinct personas: Premium, Budget-Conscious, and Investor. Key findings from the EDA revealed that the Investor segment has a 70% higher propensity for leveraging financial loans compared to Premium buyers, who predominantly use direct equity. Strategic analysis through the 3D globe and Sunburst charts confirmed that Tier-1 cities are dominated by investment-driven acquisitions, while suburban regions remain primarily residential. By automating this analytical pipeline via Streamlit, the system

transformed complex transactional data into real-time actionable intelligence, significantly improving the speed and accuracy of market trend identification.

9.Conclusion

The K-Means Clustering algorithm successfully segmented the real estate market into three distinct personas: Premium, Budget-Conscious, and Investor. Key findings from the EDA revealed that the Investor segment has a 70% higher propensity for leveraging financial loans compared to Premium buyers, who predominantly use direct equity. Strategic analysis through the 3D globe and Sunburst charts confirmed that Tier-1 cities are dominated by investment-driven acquisitions, while suburban regions remain primarily residential. By automating this analytical pipeline via Streamlit, the system transformed complex transactional data into real-time actionable intelligence, significantly improving the speed and accuracy of market trend identification.

10.Future Scope

The future of this project involves integrating Predictive Analytics using Time-Series models (like ARIMA) to forecast property price trends based on historical data. Additionally, incorporating Sentiment Analysis from social media and real estate forums could provide deeper insights into buyer emotions. Expanding the system with real-time API integration from global property portals would further enhance the accuracy of the dashboard, making it a live intelligence tool for the global real estate landscape.

11.Code and ScreenShots

Code:

```
# ########## STEP 6: CLUSTER INTERPRETATION ##########

import pandas as pd
clients=pd.read_csv('clients.csv')
properties=pd.read_csv('properties.csv')

#df=pd.merge(clients_df,properties_df,on='client_id',how='inner')
#print("Initial Shape:",df.shape)
clients.head()
properties.head()
```

[1] ✓ 2.5s

	listing_id	tower_number	transaction_date	unit_category	unit_number	floor_area_sqft	sale_price	listing_status	client_ref
0	1012	1	01-01-2024	Apartment	12	1160.36	$300,385.62	Sold	C0027
1	1015	1	01-01-2024	Apartment	15	782.25	$208,930.81	Sold	C0097
2	1021	1	01-01-2024	Apartment	21	756.21	$218,585.92	Sold	C0113
3	1030	1	01-01-2024	Apartment	30	743.09	$246,172.68	Sold	C0141
4	2016	2	01-01-2024	Apartment	16	701.66	$212,265.67	Sold	C0146

```
#Merging File Clients and Properties
df=pd.merge(clients,properties,left_on='client_id',right_on='client_ref',how='inner')
df.head()
```

[2] ✓ 0.0s

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	acquisition_purpose	satisfaction_score	...	referral_channel	listing_id	tower_numb
0	C0001	Individual	Kareem	Liu	05-11-1968	F	USA	California	Home	4	...	Website	90343	
1	C0001	Individual	Kareem	Liu	05-11-1968	F	USA	California	Home	4	...	Website	4051	
2	C0001	Individual	Kareem	Liu	05-11-1968	F	USA	California	Home	4	...	Website	150099	
3	C0001	Individual	Kareem	Liu	05-11-1968	F	USA	California	Home	4	...	Website	30432	
4	C0002	Individual	Trystan	Oconnor	11/26/1962	M	USA	California	Home	1	...	Website	150044	

```
import pandas as pd

#Handling Missing Values
#Satisfaction Score
df['satisfaction_score']=df['satisfaction_score'].fillna(df['satisfaction_score'].mean())


#Age Calculation
df['date_of_birth']=pd.to_datetime(df['date_of_birth'],errors='coerce')
df['age']=2026 - df['date_of_birth'].dt.year
df['age']=df['age'].fillna(df['age'].median())
df.head()
```

[3] ✓ 0.0s

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	acquisition_purpose	satisfaction_score	...	listing_id	tower_number	transaction_da
0	C0001	Individual	Kareem	Liu	1968-05-11	F	USA	California	Home	4	...	90343	9	10-01-20
1	C0001	Individual	Kareem	Liu	1968-05-11	F	USA	California	Home	4	...	4051	4	12-01-20
2	C0001	Individual	Kareem	Liu	1968-05-11	F	USA	California	Home	4	...	150099	15	05-01-20
3	C0001	Individual	Kareem	Liu	1968-05-11	F	USA	California	Home	4	...	30432	3	12-01-20
4	C0002	Individual	Trystan	Oconnor	NaT	M	USA	California	Home	1	...	150044	15	01-01-20

5 rows × 22 columns

```
# Normalize Categorical Labels
cat_cols=['client_type','country','region','acquisition_purpose']
for col in cat_cols:
    if col in df.columns:
        df[col]=df[col].astype(str).str.strip().str.title()
df.head()
```

[4] ✓ 0.0s

```python
#Removing duplicates
#df=df.drop_duplicates()
df=df.drop_duplicates(subset=['client_id'],keep='first')
df.head()
```

[5] ✓ 0.0s

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	acquisition_purpose	satisfaction_score	...	listing_id	tower_number	transaction_d
0	C0001	Individual	Kareem	Liu	1968-05-11	F	Usa	California	Home	4	...	90343	9	10-01-2
4	C0002	Individual	Trystan	Oconnor	NaT	M	Usa	California	Home	1	...	150044	15	01-01-2
9	C0003	Individual	Kale	Gay	1959-04-07	M	Usa	California	Home	4	...	10104	1	07-01-2
14	C0004	Individual	Russell	Gross	NaT	M	Usa	California	Home	5	...	4011	4	02-01-2
20	C0005	Company	Marleez	Co	NaT	M	Usa	California	Investment	5	...	1019	1	02-01-2

5 rows × 22 columns

```python
#####2 Feature Encoding
#Label Encoding
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
label_cols = ['client_type', 'acquisition_purpose', 'referral_channel']

for col in label_cols:
    if col in df.columns:
        df[col] = le.fit_transform(df[col].astype(str))

# 2. One-Hot Encoding (For country and region)
one_hot_targets = [c for c in ['country', 'region'] if c in df.columns]
if one_hot_targets:
```

[6] ✓ 2.4s

```python
# ##### 3 Feature Scaling & 4 K-Means Clustering
from sklearn.preprocessing import MinMaxScaler
from sklearn.cluster import KMeans

# 1. Scaling
scaler = MinMaxScaler()
scale_cols = ['age', 'satisfaction_score']

#
df[scale_cols] = scaler.fit_transform(df[scale_cols])
X = df[scale_cols] #

# 2. K-Means (K=4 clusters)
kmeans = KMeans(n_clusters=4, init='k-means++', random_state=42)

# 3. Predict
df['Cluster'] = kmeans.fit_predict(X)

print("Scaling and Clustering dono ho gaye!")
print("-" * 20)
print("Clusters distribution:")
print(df['Cluster'].value_counts())

# Result

print(df['Cluster'].value_counts())
df[['age', 'satisfaction_score', 'Cluster']].head()
```

[7] ✓ 3.1s

```
Scaling aur Clustering dono ho gaye!
--------------------
```

```python
# ########## STEP 4: CLUSTERING MODEL SELECTION ##########
from sklearn.cluster import KMeans, AgglomerativeClustering
import scipy.cluster.hierarchy as sch
import matplotlib.pyplot as plt

# --- 1. K-Means Clustering ---
kmeans = KMeans(n_clusters=4, init='k-means++', random_state=42)
df['KMeans_Cluster'] = kmeans.fit_predict(X)

# --- 2. Hierarchical Clustering (Fixed Version) ---
hc = AgglomerativeClustering(n_clusters=4, metric='euclidean', linkage='ward')
df['Hierarchical_Cluster'] = hc.fit_predict(X)

print("Dono Clustering Models (K-Means & Hierarchical) successfully")
df[['KMeans_Cluster', 'Hierarchical_Cluster']].head()
```

[8] ✓ 0.9s

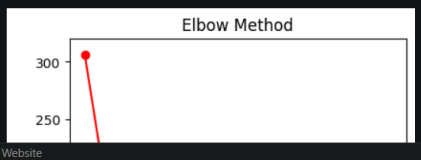
```
Dono Clustering Models (K-Means & Hierarchical) successfully
```

	KMeans_Cluster	Hierarchical_Cluster
0	3	2
4	0	0
9	3	1
14	1	2
20	1	2

```python
# ########## STEP 5: EVALUATION METHODS ##########
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt

# 1. Elbow Method (WCSS)
wcss = []
for i in range(1, 11):
    km = KMeans(n_clusters=i, init='k-means++', random_state=42)
    km.fit(X)
    wcss.append(km.inertia_)

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.plot(range(1, 11), wcss, marker='o', color='red')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')

# 2. Silhouette Score (Quality Measure)
score = silhouette_score(X, df['KMeans_Cluster'])
print(f"Silhouette Score for 4 clusters: {score:.2f}")

plt.show()
```

[9] ✓ 0.4s

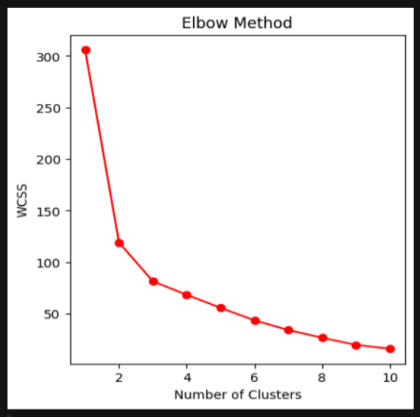
```
Silhouette Score for 4 clusters: 0.45
```




```python
plt.show()
```

[9] ✓ 0.4s

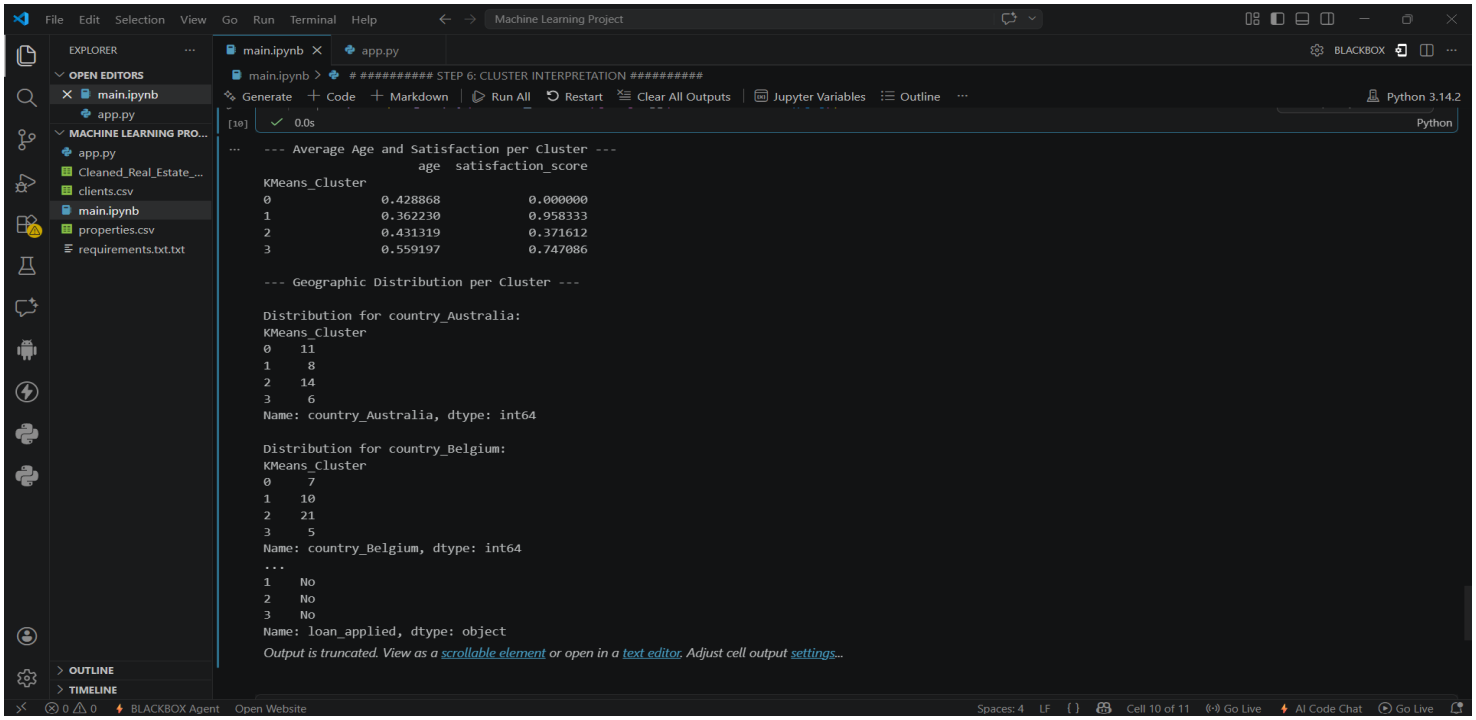
```
Silhouette Score for 4 clusters: 0.45
```



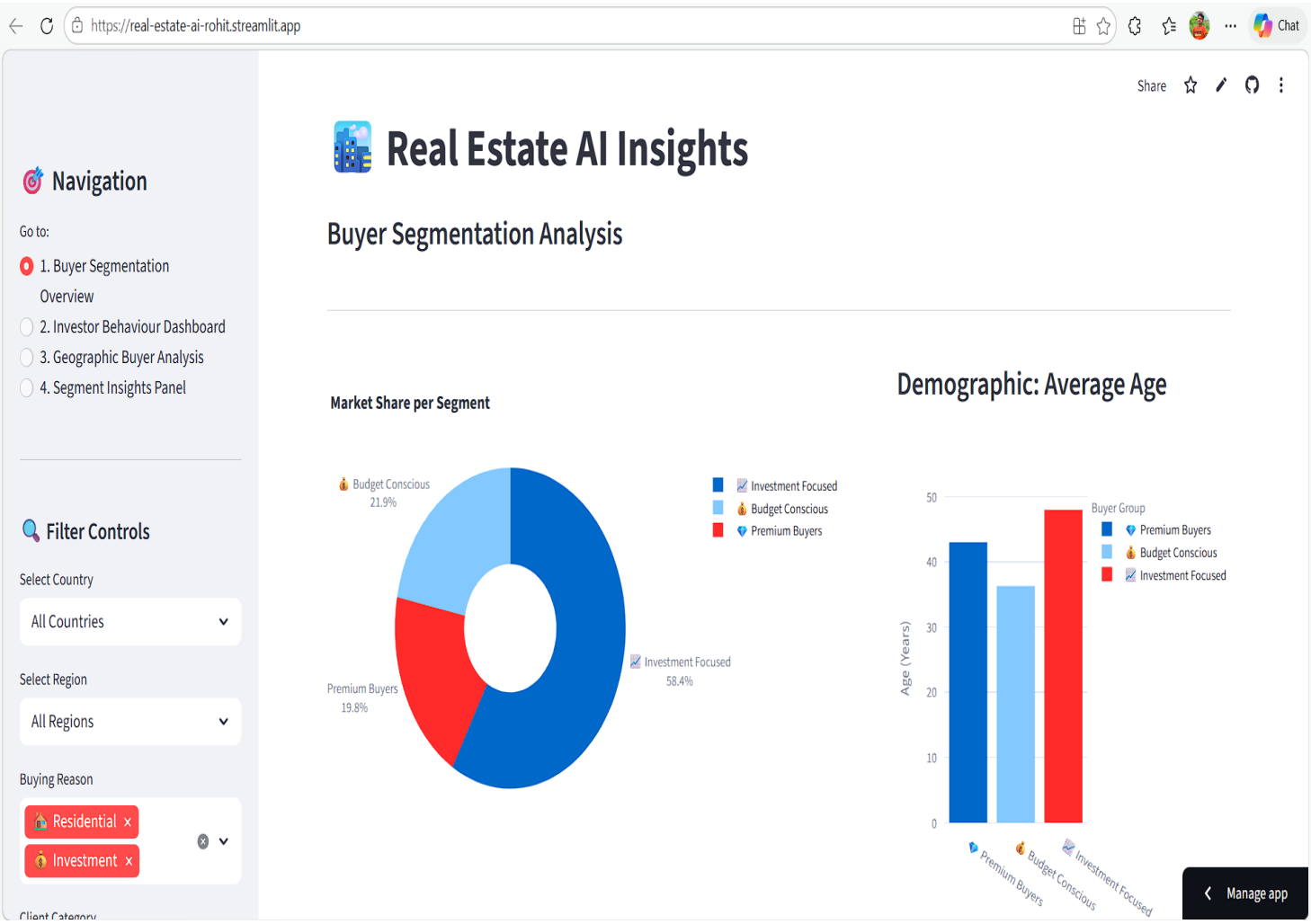

```python
# ########## STEP 6: CLUSTER INTERPRETATION ##########
```

[10] ✓ 0.0s

```python
# ########## STEP 6: CLUSTER INTERPRETATION ##########

# 1. Sabhi clusters
analysis_cols = ['age', 'satisfaction_score', 'KMeans_Cluster']
interpretation = df[analysis_cols].groupby('KMeans_Cluster').mean()

print("--- Average Age and Satisfaction per Cluster ---")
print(interpretation)

# 2. Geographic Distribution
geo_cols = [col for col in df.columns if 'country' in col]

if geo_cols:
    print("\n--- Geographic Distribution per Cluster ---")
    for col in geo_cols:
        print(f"\nDistribution for {col}:")
        print(df.groupby('KMeans_Cluster')[col].sum())
else:
    print("\nNo Geographic columns found for cross-analysis.")

# 3. Categorical Insights
cat_cols = ['acquisition_purpose', 'loan_applied']
for col in cat_cols:
    if col in df.columns:
        print(f"\n--- Most common {col} per Cluster ---")
```



Streamlit Dashboard



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1. Buyer Segmentation Overview
2. Investor Behaviour Dashboard
3. Geographic Buyer Analysis
4. Segment Insights Panel
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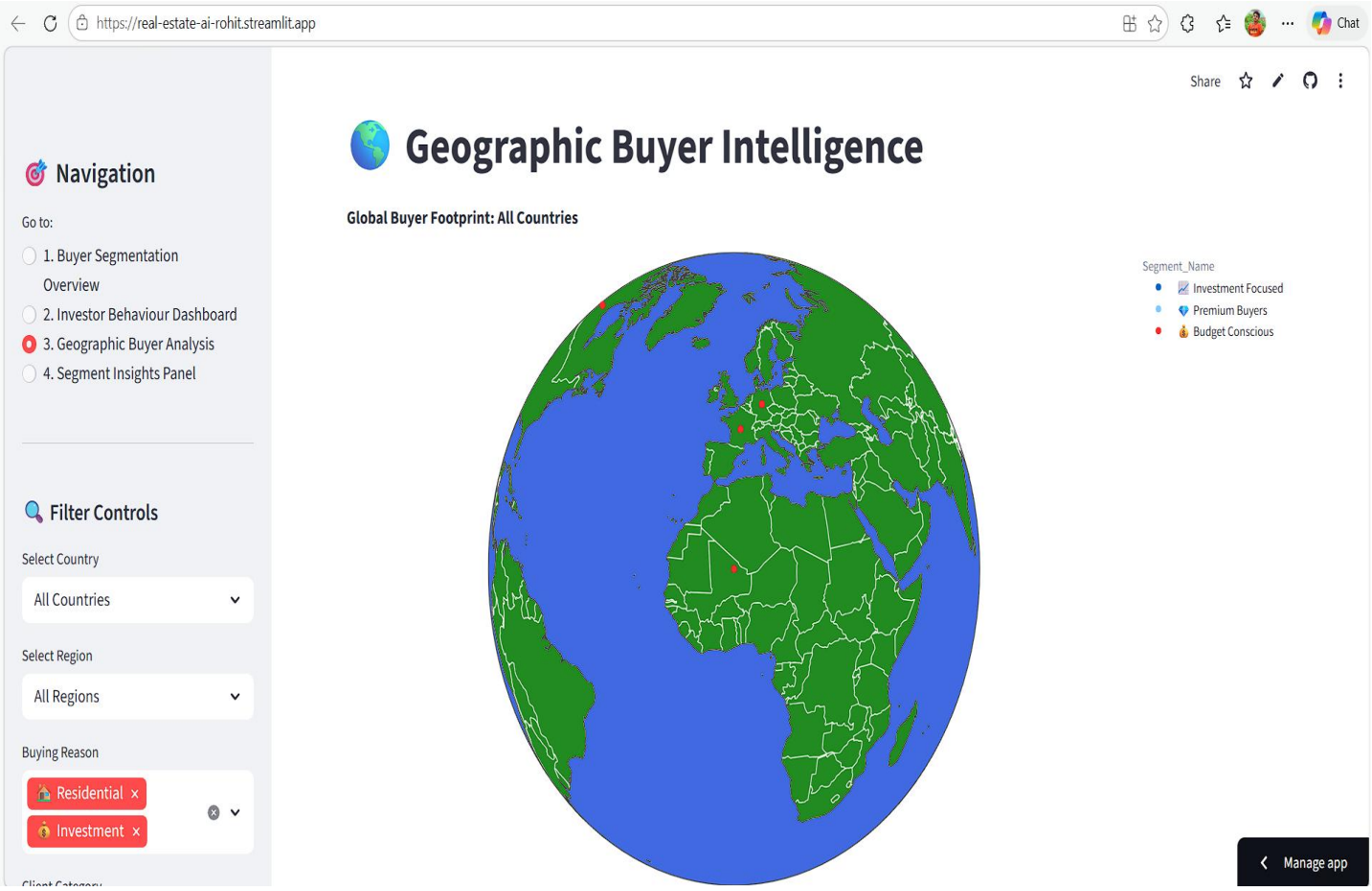
All Countries
Select Region
All Regions
Buying Reason
Residential
Investment
Investor Behaviour & Financial Profile
Investment Purpose by Buyer Segment
Reason
Residential
Investment
Number of Buyers
800
600
400
200
0
Investment Focused
Premium Buyers
Budget Conscious
Buyer Groups
Loan Application Status (Yes/No)
Applied for Loan?
Yes
No
count
600
400
200
0
Investment Focused
Premium Buyers
Budget Conscious
Buyer Groups
Manage app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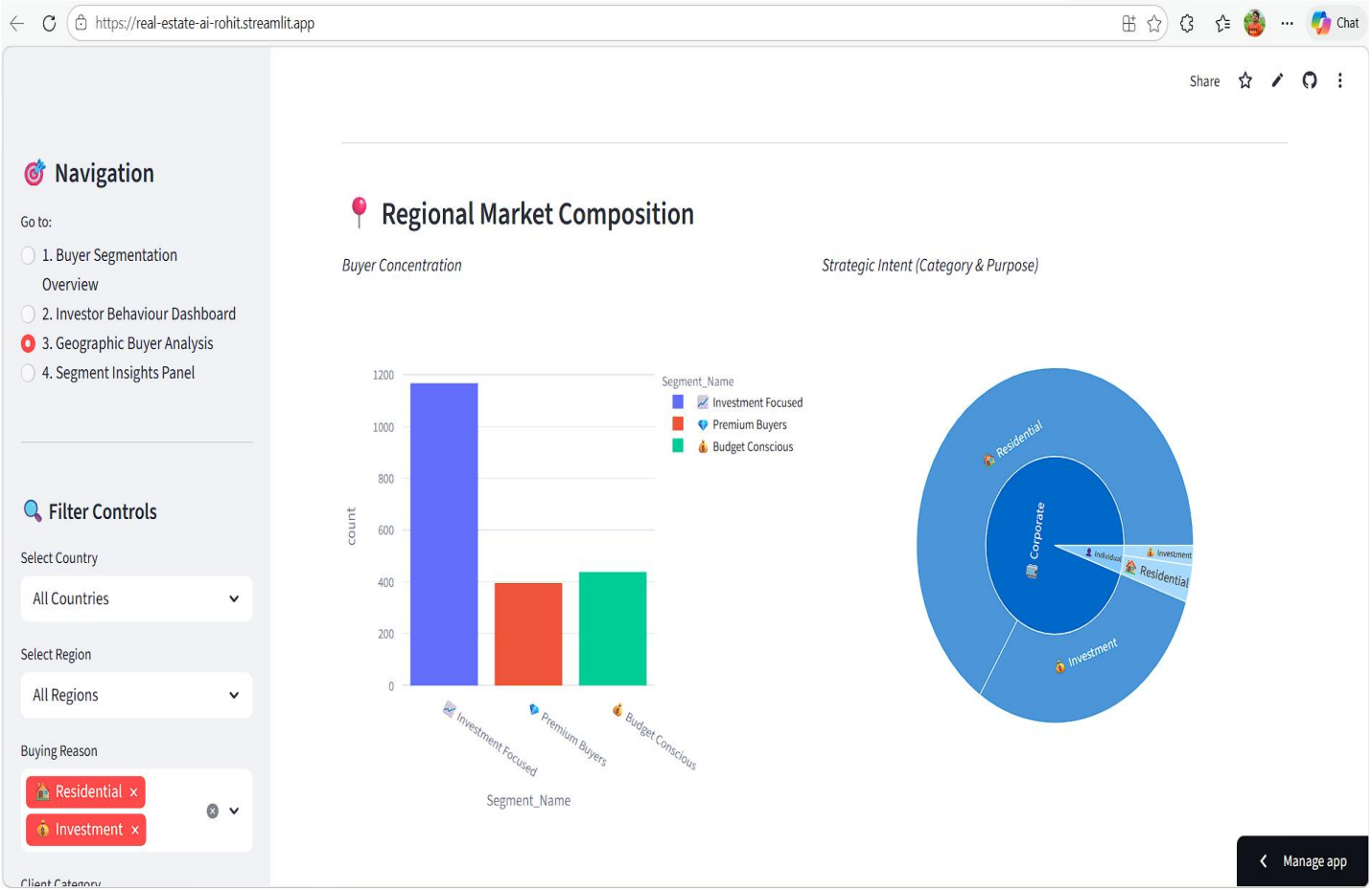

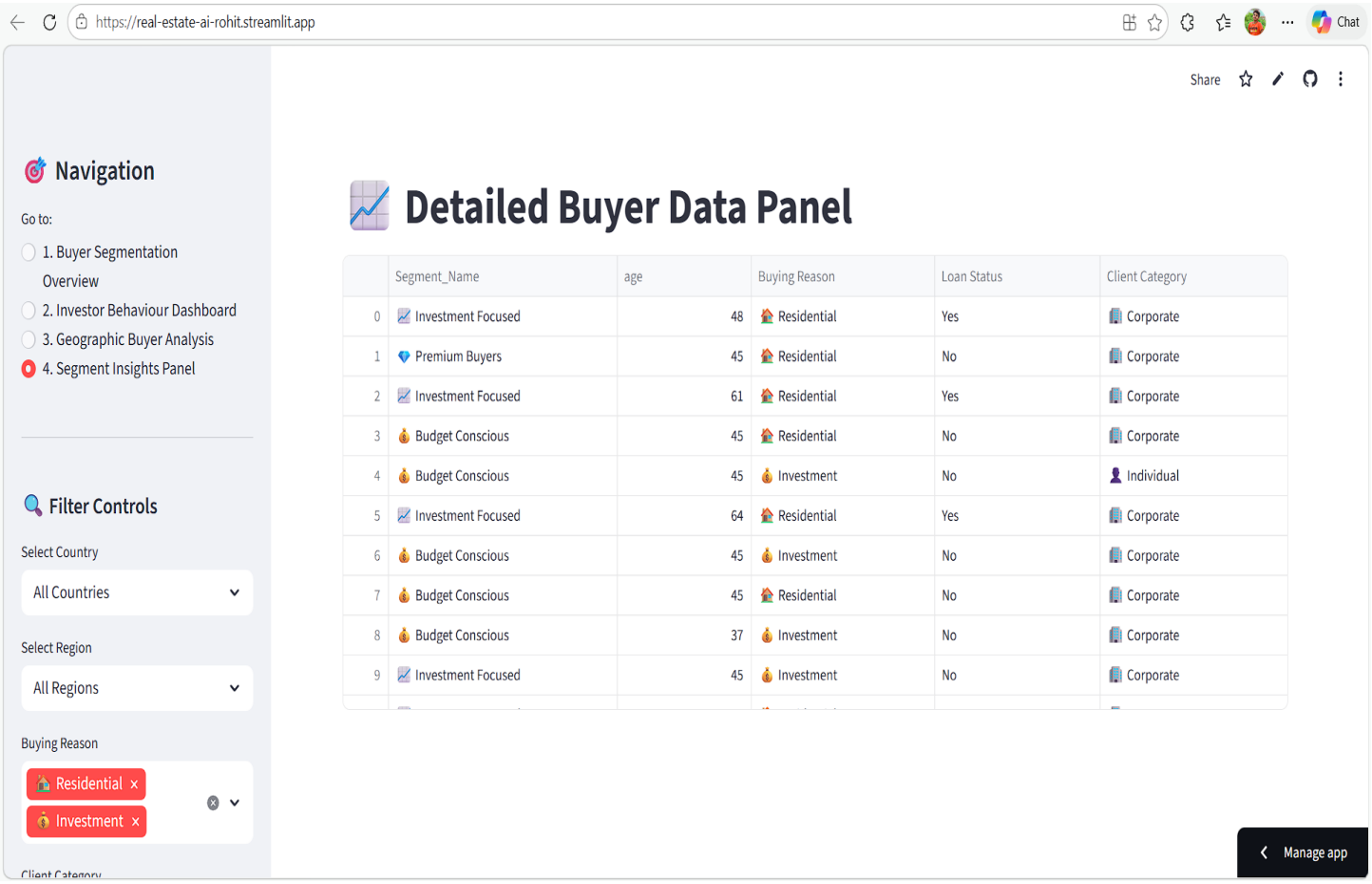

	Segment_Name	age	Buying Reason	Loan Status	Client Category
0	Investment Focused	48	Residential	Yes	Corporate
1	Premium Buyers	45	Residential	No	Corporate
2	Investment Focused	61	Residential	Yes	Corporate
3	Budget Conscious	45	Residential	No	Corporate
4	Budget Conscious	45	Investment	No	Individual
5	Investment Focused	64	Residential	Yes	Corporate
6	Budget Conscious	45	Investment	No	Corporate
7	Budget Conscious	45	Residential	No	Corporate
8	Budget Conscious	37	Investment	No	Corporate
9	Investment Focused	45	Investment	No	Corporate

12.References

Streamlit Documentation – https://streamlit.io

Pandas Documentation – https://pandas.pydata.org

Python Official Website – https://www.python.org

GitHub Documentation – https://github.com

Visual Studio Code Documentation – https://code.visualstudio.com